

NIRAV SURABHI

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Mechanical engineering student at Purdue and first-author IEEE researcher who ships production software — native iOS apps, real-time backends, and applied AI — alongside CAD, controls, and instrumented hardware.

EDUCATION

Purdue University — **B.S. First-Year Engineering** · **Intended Mechanical Engineering** · AUG 2026 — MAY 2030
Coursework: ENGR 133, MA 265 Linear Algebra, PHYS 172 Modern Mechanics, SCLA 102 · Purdue Formula SAE (onboarding, Fall 2026).

Independence High School — **Loudoun County, VA** · GRADUATED JUN 2026

Coursework: AP Computer Science A, AP Physics C, Multivariable Calculus.

EXPERIENCE

SpareMe — **Founder & Lead Engineer** · **getspareme.com** · JUL 2026 — PRESENT

- Shipped a native SwiftUI iPhone app to the App Store that delegates phone calls to an AI agent — it dials U.S. businesses, works phone trees, waits on hold, and returns a written outcome; released after **13 TestFlight builds** and an invite beta.
- Engineered the Node/TypeScript backend on **Twilio ConversationRelay** for real-time speech, with OpenAI reasoning, verified caller ID, encrypted PostgreSQL case state, APNs notifications, signed StoreKit verification, and fail-closed kill switches and rate limits.

Tech: Swift/SwiftUI · TypeScript · Node.js · Twilio · OpenAI · PostgreSQL · StoreKit · APNs

Khali — **Co-Founder & Lead Engineer** · **khali.info** · 2025 — PRESENT

- Shipped the native **SwiftUI iOS** app and the **Expo / React Native** app for a short-form news feed that combines opposing sources one claim at a time; launched on the App Store and applied to **Y Combinator** Spring 2026.
- Built the Node + Python backend that ingests sources, clusters events into stories, and generates the clips and narration, plus an OpenAI assistant that surfaces the evidence behind each claim.

Tech: Swift/SwiftUI · React Native · Expo · TypeScript · Node.js · Python · OpenAI · Supabase

I3 Simulations — **CAD Intern** · JUN 2025 — MAY 2026

- Produced 3D models and haptic-feedback prototypes for medical-training simulators.

The Bike Lane — **Bike Mechanic (intern → part-time)** · JUN 2021 — AUG 2025

- Assembled bicycles from the frame up and ran the final safety inspection before customer delivery.

Ember Learning — **AI Engineer & Software Intern** · **formerly DeAP Learning Labs** · FEB 2021 — 2022

- Wrote the LLM prompts behind AP-tutor chatbots, built automated grading of AP DBQ/FRQ responses against the official rubrics, indexed lecture transcripts in a vector database for timestamp search, and led the company website. **Team secured a Microsoft and OpenAI grant; company valued at \$1M+.**

RESEARCH & PUBLICATIONS

Passive Vortex Propulsion — **Published** · **IEEE Xplore, Document 11567580** · 2024 — 2025 · **FIRST AUTHOR**

- Built a self-made wind tunnel and test rig, then soldered and calibrated an Arduino load-cell system to measure thrust directly.
- Ran CFD in Fusion 360 and analyzed 20 trials in Python, measuring **3.1 N mean net thrust (p < 0.001)** — the first experimental evidence of passive vortex propulsion in air. Earned the **AIAA Aerospace Award** at RSEF with an invitation to NASA Goddard; presented at EngiTek 2025.

Tech: Arduino · C++ · load cells · Fusion 360 · CFD · Python · statistical analysis

PROJECTS

Oh — **Creator** · **github.com/21J3phy/Oh** · 2026 — PRESENT

- Built shared memory for AI coding agents: hook-driven session capture, local secret scrubbing, pgvector retrieval, and an MCP tool that answers why a change was made instead of interrupting its author.

FTC Robotics, Absolute Zero #12096 — **Captain & Lead Programmer** · 2022 — 2026

- Led the team for **four seasons**, CADding and building every part in Fusion 360 and writing autonomous and tele-op code in Java with PID control, sensor fusion, and odometry; advanced to the Chesapeake championship.

Motor-Based Resistance System — **Portable battery-powered strength trainer** · 2025 · RSEF · **FIRST AUTHOR**

- Designed and built three prototypes in which a 188:1 geared motor produced a repeatable PWM-controlled force curve (**R² = 0.993**) at a \$214.95 prototype cost.

Chad vs. Chud — **Co-Founder & Lead Engineer** · **real-time PvP web app** · 2026 · SUNSET

- Built a Next.js webcam client, Socket.IO/WebRTC matchmaking, and a FastAPI service running a fine-tuned **ResNet-18 + ArcFace** ensemble; scaled to **10,000+ users, 100+ concurrent**.

TECHNICAL SKILLS

LANGUAGES	Python, Java, C++, Swift, TypeScript, JavaScript, SQL
SOFTWARE	SwiftUI, React Native, Expo, Next.js, Node.js, FastAPI, PostgreSQL, Supabase, REST, WebSockets, Git, CI/CD, Vercel
AI / ML	PyTorch, computer vision, LLM integration, vector search, agent architecture, prompt engineering, evaluation
HARDWARE	Fusion 360, CAD, FEA, CFD, 3D printing, Arduino, load cells, soldering, PID, PWM motor control, sensor fusion, odometry, statistical analysis

HONORS & AWARDS

- AIAA Aerospace Award, Regional Science & Engineering Fair — special honor, with invitation to NASA Goddard · Presidential Volunteer Service Award, Silver
- Central Virginia Governor's School for Science & Technology — Mathematics Scholar, 2024 · Captain, Science Olympiad · 1st place, Aguado Guitar Competition · All-Virginia Guitar Ensemble